DR. SINTER LAB Jr. Series Spark Plasma Sintering Systems

Compact Desktop & Laboratory R&D Materials Synthesis Platform - Official System Brochure

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OEM: FUJI ELECTRONIC INDUSTRIAL CO., LTD. (JAPAN) | SPS TECHNOLOGY DIVISION (fdc.co.jp/sps)
INDIAN TECHNICAL SALES & DEPOT SUPPORT: SYINCO TECHNOLOGIES, HYDERABAD, TELANGANA, INDIA

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VERIFIED TECHNICAL SPECIFICATIONS & OPERATING PARAMETERS:

* Technology: Spark Plasma Sintering (SPS) / Pulsed Electric Current Sintering (PECS)
* Maximum Pressure: 20 kN to 60 kN (2,040 kgf to 6,120 kgf) via AC Servo Motor
* Minimum Pressure: 0.5 kN (51 kgf) for micro-load fragile green compacts
* Maximum Sintering Temperature: 2,500 deg C with Optical Pyrometer / Thermocouple
* Sintering DC Pulse Generator: 1,000 A to 3,000 A (Input: AC 200V, 3-Phase, 50/60 Hz)
* Pulse Timing Control: ON: 1 to 999 ms / OFF: 1 to 99 ms (continuous variable pulse)
* Vacuum Chamber: Water-Cooled Stainless Steel Chamber (Bore: 200 mm to 259 mm)
* Electrode System: Special Sealed Water-Cooled Electrode Assembly
* Displacement Sensor: Digital Linear Gauge for Real-Time Shrinkage Monitoring

CANONICAL MODEL CONFIGURATIONS & FORCE ENVELOPES:

> Model MS-1: 20 kN Force, 1,000 A Current, 259 mm Chamber Bore, 50 mm Stroke
> Model 212Lx: 20 kN Force, 1,000 A Current, 200 mm Chamber Bore, 50 mm Stroke
> Model 322Lx: 30 kN Force, 2,000 A Current, 200 mm Chamber Bore, 150 mm Stroke
> Model 632Lx: 60 kN Force, 3,000 A Current, 200 mm Chamber Bore, 150 mm Stroke

SCOPE OF DELIVERY & SYINCO LOCAL SUPPORT ADVANTAGE:

- Complete OEM Furnace Body, Chamber, Electrode Assembly & DC Pulse Power Supply
- In-Country INR Supply & Customs Clearance Handled Ex-Hyderabad Central Logistics Depot
- Certified Field Installation, Commissioning, Calibration & Preventive Maintenance (AMC)
- Dedicated Inventory of High-Purity Graphite Tooling (Dies & Punches), Thermocouples & Seals

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